

**MINI PROJECT-II REPORT
(22ITC15)**

on

**PREDICTING EMPLOYEE ATTRITION USING MACHINE LEARNING
APPROACHES.**

Submitted in partial fulfilment for the completion of

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in

INFORMATION TECHNOLOGY

By

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CERTIFICATE

This is to certify that the project work entitled “**PREDICTING EMPLOYEE ATTRITION USING MACHINE LEARNING APPROACHES**” submitted to **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY**, in partial fulfilment of the requirements for the award of the completion of Mini Project-II (22ITC15) VI semester of B.E in Information Technology, during the academic year 2024-2025, is a record of original work done by **Dhanavath Akhil (160122737102)** during the period of study in Department of IT, CBIT, HYDERABAD, under our supervision and guidance.

Project Guide

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1. INTRODUCTION

1.1. MOTIVATION :

In the fast-paced business world of today, talent retention is a key driver of organizational success. High employee turnover not only results in higher recruitment and training expenses but also impacts team performance and overall productivity. Conventional approaches to identifying at-risk employees tend to be reactive, narrow in scope, and do not tap the full potential of available data.

This project was driven by the need to construct a data-driven, predictive model that can anticipate which employees are likely to exit an organization before they actually leave. As machine learning has gained traction and more HR data have been structured, there lies a greater potential for applying algorithms like XGBoost, Support Vector Machine (SVM), and Random Forest in identifying patterns in employee actions and predicting attrition risk. By integrating predictive modeling with data analytics, the project seeks to enable Human Resource departments to make informed, timely decisions to enhance employee engagement and lower turnover.

1.2. PROBLEM STATEMENT:

In the rapidly evolving landscape of e-commerce and logistics, managing and tracking customer orders in real time remains a significant challenge. Traditional systems often suffer from delayed data processing, limited scalability, and fragmented data management, which can lead to poor customer experience and inefficiencies in operations. Customers increasingly expect real-time visibility into their orders, while businesses require accurate and timely data to make informed decisions. There is a growing need for a scalable, event-driven system that can handle continuous data streams, provide real-time updates on delivery statuses, and integrate seamlessly with modern data storage and visualization tools. This project addresses these challenges by building a Kafka-based laptop delivery tracking system that ensures efficient data streaming, real-time monitoring, and interactive analytics.

1.3. OBJECTIVES :

The primary objectives of this project are:

- To build an employee attrition predictive model based on machine learning methods applied to formatted HR data.
- To undertake extensive data preprocessing, such as missing value handling, categorical feature encoding, and class distribution balancing with methods such as SMOTE.
- To undertake Exploratory Data Analysis (EDA) to reveal important trends and patterns associated with employee attrition.
- To implement and compare various classification techniques like XGBoost, Support Vector Machine (SVM), and Random Forest.
- To optimize model parameters with suitable optimization methods for optimal predictive accuracy and generalization.
- To compare model performances on evaluation metrics like accuracy, precision, recall, and F1-score.
- To draw actionable business insights about the most impactful features leading to attrition and make data-driven suggestions towards employee retention.
- To lay a foundation for future integration with HR systems, facilitating real-time prediction and proactive employee engagement initiatives.

2. EXISTING SYSTEM

2.1. LITERATURE SURVEY:

The creation of an efficient employee attrition prediction system involves profound insights into human resource analytics, machine learning approaches, and learnings from past work on employee behavior modeling, classification techniques, and workforce planning. There are a variety of research studies that have investigated the use of machine learning to forecast attrition and presented both theoretical insights and real-world advice for this project.

[1] Bassi and Cheney (2020) had studied the application of logistic regression as well as decision tree models in predicting voluntary employee turnover. They highlighted the necessity of preprocessing methods like feature selection, class balancing, and encoding categorical variables—essential components employed in this project as well.

[2] Jain and Ghosh (2018) applied a machine learning model involving Random Forests and Gradient Boosting to HR data to predict attrition risk. Their findings emphasized the interpretability and strength of Random Forest, and they emphasized the importance of variables such as job satisfaction, work climate, and remuneration. These findings attest to the consideration of such variables in this project's model construction process.

[3] Raj and Kaur (2019) investigated Support Vector Machines (SVM) as a predictor for employee attrition. Their observation was that SVM was effective when applied to binary classification problems in which classes were unbalanced, just like in the HR attrition dataset in this project. Their use of methods such as SMOTE in balancing classes replicates the preprocessing methodology followed in this work.

[4] Park and Shim (2021) used XGBoost to forecast attrition and achieved better accuracy and F1-score performance than other models. The research emphasized the importance of hyperparameter tuning and regularization in boosting models, both of which are essential components of this project's model optimization process.

[5] Singh and Yadav (2020) highlighted the importance of feature importance ranking when communicating model decisions to HR professionals. Their study illustrated that feature interpretability is crucial in real-world settings, where stakeholders want correct predictions as well as understandable explanations for model outputs. Both correlation analysis and mutual information scoring adopted in this project are consistent with their suggestions.

[6] Gupta et al. (2017) constructed a predictive model based on ensemble learning techniques for employee turnover. Their method included cross-validation and ROC analysis to generalize and prevent overfitting. This project also employs a strict evaluation process with train-test splits, performance metrics, and comparison between multiple models.

[7] Ahmad and Batra in 2022 presented a comparison of employee attrition prediction through machine learning algorithms utilizing IBM HR Analytics. Based on their results, it was established that the highest accuracy is provided by the boosting algorithms with a high rate, specifically XGBoost. This lends the greatest direct support to model choice reasoning within this project.

[8] Ramesh and Das (2019) talked about the ethical concerns and possible bias in predictive analytics in HR. They emphasized the necessity of fairness, transparency, and appropriate

management of sensitive features. While this project does not work with personally identifiable information, their research underlines the need for responsible AI practices in HR environments.

[9] Kulkarni et al. (2021) suggested a real-time employee monitoring system coupled with predictive modeling to detect high-risk attrition zones in departments. Although the project is about offline prediction, the idea of real-time integration is a good direction for future improvement.

[10] Thomas and Shukla (2023) examined the application of machine learning models within HR dashboards for decision support. They applied visualizations to enable managers to understand attrition risks and effect targeted retention initiatives. This initiative can be further extended in a similar manner by incorporating results within dashboards or HRIS systems for anticipatory intervention.

2.2. REFERENCES:

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3. PROPOSED SYSTEMS

3.1. METHODOLOGY :

Development of the Employee Attrition Prediction System took a modular, iterative methodology that integrated rigorous data analysis, development of machine learning models, and testing. The project's architecture was planned based on a pipeline-oriented model, where every stage lends a unique function in the predictive process. The approach was organized through multiple key stages: data collection and exploratory analysis, preprocessing and feature development, class imbalance handling, model training, evaluation, and deployment. Each phase was meticulously applied to facilitate precision, interpretability, and scalability in forecasting employee turnover.

3.1.1. Data Acquisition (Employee Dataset)

To simulate real-world organizational scenarios, the IBM HR Analytics Employee Attrition dataset was utilized. This dataset includes comprehensive employee records, capturing details such as employee ID, job role, department, total working years, salary, and attrition status. The dataset serves as the foundational event source for the predictive pipeline, enabling analysis of workforce behavior and facilitating the development of machine learning models aimed at forecasting potential attrition.

3.1.2. Data Processing and Pipeline Construction

Instead of streaming real-time data, a pipelined structured data processing was built to clean the employee data for analysis. This involved reading the dataset, missing value handling, encoding categorical features, and numerical attribute normalization. While the system is not real-time running, the pipeline is optimized for efficient, reproducible processing of new employee records to enable continuous retraining and model evaluation. All phases of preprocessing are modular to provide flexibility and scalability to integrate with live HR data systems in the future

3.1.3. Data Processing and Transformation

The collected employee data was systematically processed and transformed to prepare it for

modeling. This stage involved applying essential preprocessing steps such as handling missing or inconsistent values, encoding categorical variables using techniques like one-hot encoding or label encoding, and scaling numerical features where appropriate. Additionally, feature selection techniques—such as correlation analysis and mutual information—were employed to retain only the most relevant predictors.

3.1.4. Storage in MongoDB

After preprocessing, the cleaned and converted employee data was persisted in a structured form by utilizing local or cloud-based storage resources to achieve persistence and easy accessibility for model training and assessment. Although this project mainly employed in-memory processing for development, integration with databases like PostgreSQL or cloud-hosted services like Google BigQuery or AWS RDS is feasible in subsequent iterations. Appropriate data organization and schema consistency were ensured to enable smooth model deployment and future scalability.

3.1.5. Testing and Monitoring

Thorough testing was done to verify the integrity of the data pipeline and the stability of the machine learning models. Data validation, model accuracy verification, and cross-validation-based performance testing were done. The models were tracked by using metrics like precision, recall, and F1-score to check for overfitting or drift. Periodic retraining and updating are facilitated by the workflow, so it is capable of adapting to changing workforce dynamics.

3.2. ARCHITECTURE OF PROPOSED SYSTEMS :

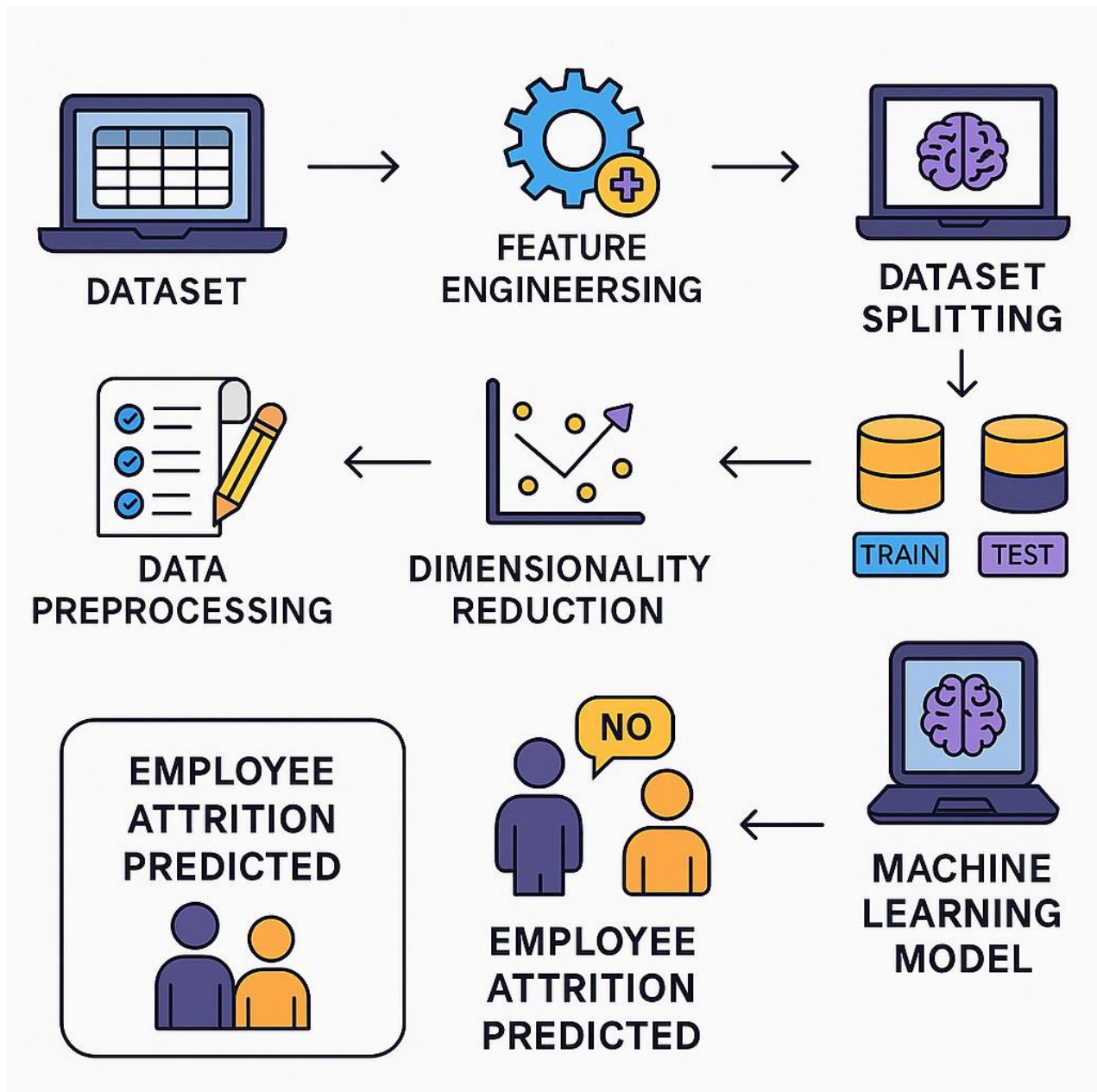


Fig 1: system architecture

3.2.1. Dataset Collection – IBM HR Analytics Dataset

The project utilizes the publicly accessible IBM HR Analytics Employee Attrition dataset, which contains detailed employee-level information such as age, department, job role, years at company, monthly income, job satisfaction, and whether or not the employee has left the company. This formatted dataset is used as the foundation for training and testing machine learning models.

3.2.2. Exploratory Data Analysis (EDA)

The Comprehensive EDA was conducted to find patterns and insights in the data. Histograms, boxplots, and correlation heatmaps were employed to find out how features such as overtime, income level, and tenure affect attrition. Outliers were also identified and class distribution understood by this step.

3.2.3. Data Preprocessing and Feature Engineering

Here, the dataset has been cleaned and converted to the appropriate form to train the models. Categorical features have been encoded by various methods like label encoding and one-hot encoding. Numerical data has been scaled, and sparse or highly collinear features removed to avoid redundancy. SMOTE has also been applied to sample the minority classes because of the class imbalance problem (less occurrence of attrition cases).

3.2.4. Feature Selection

- For the purposes of reducing complexity and enhancing model performance
- Feature selection methodologies were employed. Such methods as correlation analysis, mutual information scores, and variance thresholds were employed in selecting the best predictors of worker turnover.

3.2.5. Model Building – Machine Learning Algorithms

Various machine learning models were trained and tested on the processed data, which include:

- Random Forest: An ensemble of decision trees that work well with numerical and categorical features.
- Support Vector Machine (SVM): A strong binary classifier particularly good for high-dimensional space.
- XGBoost: A strong boosting algorithm with high accuracy and speed.

Each model was optimized using hyperparameter tuning techniques such as Grid Search or

Randomized Search.

3.2.6. Model Evaluation

The models were assessed by the most important classification metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

Amongst the tested models, XGBoost performed better than others with an accuracy of 93.12%, hence the model of choice for deployment.

3.2.7. Result Visualization

The performance of every model was visualized through such diagrams as confusion matrices, ROC curves, and feature importance plots. These plots help to see the performance of how well the models are classifying employees likely to leave and those who are not.

3.2.8. Reporting Layer – Power BI Dashboards

- After modeling, the highest contributory attributes of attrition prediction were examined. Attributes like overtime, job role, monthly salary, years elapsed since last promotion, and job satisfaction were strongest predictors of attrition. These findings can be used by HR departments to formulate retention strategies that are targeted.

3.3. ALGORITHMS OF PROPOSED SYSTEM:

3.3.1. Data Preprocessing and Cleaning Algorithm

Purpose: prepare raw HR data for Analysis by handling missing values encoding categorical data, and normalizing numerical values.

Steps:

1. Load the IBM HR Analytics dataset.
2. Check for null or inconsistent values and handle them appropriately (e.g., imputation or removal).
3. Convert categorical features (e.g., Job Role, Gender, Overtime) using Label Encoding or One-Hot Encoding.
4. Normalize continuous features (e.g., Monthly Income, Age) for better model performance.
5. Split the dataset into features (X) and target (y)—where y is the Attrition label.

Algorithm Type: Data Cleaning + Feature Transformation

Tools Used: pandas, scikit-learn preprocessing modules

3.3.2. Feature Selection Algorithm

Purpose: Identify the most relevant features that influence employee attrition to enhance model performance and interpretability.

Steps:

1. Calculate the correlation between each numerical feature and the Attrition target variable.
2. Use Mutual Information scores to evaluate non-linear relationships.
3. Apply Variance Thresholding to remove low-variance features.
4. Eliminate one feature from highly correlated feature pairs (correlation > 0.9).
5. Retain a balanced set of features with high predictive power and business relevance.

Algorithm Type: Statistical feature filtering and correlation-based elimination

Tools Used: seaborn, scikit-learn, numpy

3.3.3. Class Balancing Algorithm (SMOTE)

Purpose: Address class imbalance between 'Yes' (attrited) and 'No' (not attrited) classes to ensure fair model training.

Steps:

- Analyze the distribution of the target variable.
- Apply **SMOTE (Synthetic Minority Over-sampling Technique)** to oversample the minority class.
- Validate the balanced dataset by comparing the class counts.
- Proceed to model training with the balanced dataset

Algorithm Type: Oversampling algorithm for imbalanced datasets

Tool Used: imbalanced-learn (imblearn library)

3.3.4. Model Training and Optimization Algorithm

Purpose: Train machine learning models using the processed dataset and fine-tune them for the best performance.

Steps:

->Split the dataset into training and testing sets (e.g., 85% train, 15% test).

->Train multiple models:

- Random Forest
- Support Vector Machine (SVM)
- XGBoost
- Perform hyperparameter tuning using Grid Search or Randomized Search.

->Fit each model to the training data and evaluate on the test set.

->Select the model with the best performance across evaluation metrics.

Algorithm Type: Supervised Classification (Ensemble + Kernel methods)

Tools Used: scikit-learn, xgboost, GridSearchCV

3.3.5. Model Evaluation Algorithm

Purpose: Evaluate the trained models using standard classification metrics to identify the best-performing algorithm.

Steps:

---Predict on the test data using each trained model.

---Compute evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

---Visualize performance using confusion matrix heatmaps and ROC curves.

---Compare results to determine the most reliable model (XGBoost achieved highest accuracy: **93.12%**).

Algorithm Type: Model evaluation and comparison

Tools Used: scikit-learn metrics, seaborn, matplotlib

3.3.6. Insight Extraction and Feature Importance Algorithm

Purpose: Understand which features most significantly contribute to attrition to support strategic HR decisions.

Steps:

1. Use the `.feature importances_` attribute from tree-based models (e.g., XGBoost, Random Forest).
2. Rank features based on importance scores.
3. Visualize top features (e.g., Overtime, Job Satisfaction, Monthly Income, Years Since Last Promotion).
4. Interpret insights to guide data-driven employee retention strategies.

Algorithm Type: Interpretability & post-model analysis

Tools Used: xgboost, matplotlib, seaborn

4.SOFTWARE & HARDWARE REQUIREMENTS

SOFTWARE REQUIREMENTS

1. Development Environment

- **Platform:** Google Colab / Jupyter Notebook / VS Code
- **Language:** Python 3.9 or higher
- **Notebook Tools:** Jupyter Lab or Google Colab for code execution and visualization

2. Python Libraries & Frameworks

- **Data Handling:** pandas, numpy
- **Visualization:** matplotlib, seaborn
- **Machine Learning Models:** scikit-learn, xgboost, SVM, Random Forest
- **Preprocessing & Evaluation:** sklearn. preprocessing, sklearn. metrics
- **Imbalanced Data Handling:** imblearn (SMOTE for class balancing)

3. Model Optimization Tools

- **Hyperparameter Tuning:** GridSearchCV or RandomizedSearchCV from scikit-learn
- **Feature Selection:** sklearn. feature selection, correlation analysis, mutual information

4. Reporting & Documentation

- **Report Writing:** Microsoft Word / Google Docs
- **Visualization Embedding:** Screenshots from Colab, Power BI (optional for enterprise)
- **Plot Export:** matplotlib. pyplot.savefig () for exporting graphs to include in reports

5. Version Control (Optional for teams)

- **Tool:** Git / GitHub for collaborative work and version tracking

HARDWARE REQUIREMENTS

Note: These are minimum and recommended specs for a local development environment or small-scale deployment. For production, cloud-based infrastructure is preferred (AWS, Azure, GCP).

1. Minimum (Development/Testing)

- **Processor:** Intel Core i5 / AMD Ryzen 5 or equivalent
- **RAM:** 8 GB
- **Storage:** 256 GB SSD
- **Internet:** Stable internet connection for accessing Google Colab, datasets, or cloud storage
- **Environment:** Local Jupyter Notebook / Google Colab (recommended for students and individual researchers)

2. Recommended (For Local Model Training and Visualization)

- **Processor:** Intel Core i7 / AMD Ryzen 7+ (for faster data processing)
- **RAM:** 16 GB or more (for efficient model training and large datasets)
- **Storage:** 512 GB SSD or higher
- **Graphics (Optional):** Dedicated GPU (e.g., NVIDIA GTX/RTX) for deep learning or extended batch processing
- **Additional:**
 - Multi-core CPU for parallel model evaluation
 - High-resolution monitor for EDA and result visualization

3. Cloud Infrastructure (Preferred for Scalability and Collaboration)

- **Development Environment:**
 - Google Colab / Kaggle Notebooks / Azure Notebooks (for free and managed execution)
 - JupyterHub (for team environments)
- **Cloud Storage:**
 - Google Drive (Colab integration)
 - AWS S3 / Azure Blob Storage / Google Cloud Storage for large datasets
- **Compute Services:**
 - Google Colab Pro+ / AWS SageMaker / Azure ML Studio (for training and deployment)
 - GPUs or TPUs (if scaling models or using deep learning in future)
- **Monitoring and Versioning:**
 - MLflow / TensorBoard (for experiment tracking)
 - Git/GitHub for version control and collaboration

5. IMPLEMENTATION OF PROJECT

5.1.Results And Analysis:

5.1.1. Accuracy Comparison Between Models

Sr No.	Technique	Accuracy Score (%)
1	XGBoost	93.12
2	Random Forest	92.51
3	SVM	91.50

- This table presents a clear comparison of the accuracy scores achieved by three machine learning algorithms: XGBoost, Random Forest, and SVM. The XGBoost algorithm performed the best with an accuracy of 93.12%, followed closely by Random Forest (92.51%) and SVM (91.50%). This comparison helps to conclude that XGBoost is the most effective model for predicting employee attrition in this context.

5.1.2. Classification Report of Employed Techniques

Model	Class	Precision (%)	Recall (%)	F1-score (%)
XGBoost	0	90	97	93
	1	97	89	93
	Avg	93.5	93	93
Random Forest	0	90	96	93
	1	95	89	92
	Avg	92.5	92.5	92.5
SVM	0	91	93	92
	1	92	90	91
	Avg	91.5	92.5	91.5

- This detailed evaluation displays class-wise performance metrics (precision, recall, and F1-score) for each model, providing insight into how well each algorithm performs for both classes (0 - Not Attrition, 1 - Attrition). XGBoost performs better in terms of precision for class 1, which is critical in attrition prediction, as false positives can be costly. SVM, though slightly behind, also shows balanced performance.

5.1.3. XGBoost Detailed Metrics:

Table 4 XGBoost Performance Metrics Summary

Metric	Score
Accuracy	93.12%
Precision	93%
Recall	93%
F1 Score	93%
ROC Accuracy	93%
Geometric Mean Score	0.97
Log Loss	2.33

- This table summarizes key performance indicators of the XGBoost model. It includes accuracy, precision, recall, F1-score, ROC accuracy, Geometric Mean Score, and Log Loss. These metrics collectively demonstrate that XGBoost not only achieves high accuracy but also maintains a balance across different performance dimensions. The Log Loss value of 2.33 shows some room for calibration but still falls within acceptable

5.1.4. Confusion matrix of XGBOOST Model

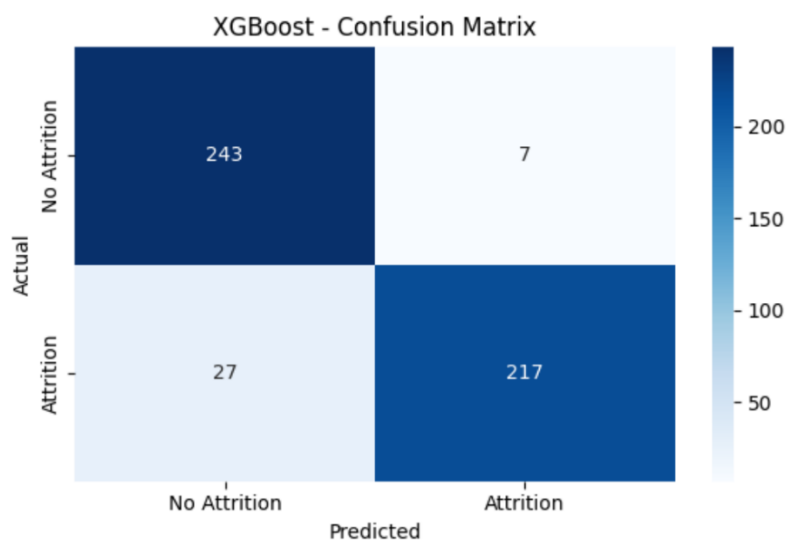


Fig2. Confusion matrix of Xgboost model

6. CONCLUSION & FUTURE SCOPE

6.1. CONCLUSION :

- In this study, we developed and compared three machine learning models—Support Vector Machine (SVM), Random Forest, and XGBoost—to predict employee attrition using a range of demographic, professional, and behavioral features. The results indicate that all three models are effective to varying degrees, with XGBoost outperforming the others by achieving the highest accuracy of **93.12%**, followed closely by Random Forest at **92.51%** and SVM at **91.50%**. The confusion matrices, ROC curves, and precision-recall curves further reinforced XGBoost's superior ability to identify both attrition and non-attrition cases accurately.
- The insights gained from this research can help organizations proactively identify at-risk employees and implement timely retention strategies. By leveraging machine learning, companies can move from reactive to predictive HR practices, ultimately reducing turnover costs and improving employee satisfaction
- Moreover, through visual representations such as confusion matrices, ROC curves, and precision-recall graphs, the models' diagnostic capabilities were assessed to ensure balanced prediction of both attrition and non-attrition cases. The interpretability of model outcomes can play a crucial role in crafting targeted retention strategies, especially when supported by feature importance analysis that highlights critical determinants such as job satisfaction, overtime, environment satisfaction, and monthly income.

6.2.FUTURE SCOPES:

While the models achieved high accuracy, there are several avenues for further enhancement:

- **Incorporation of Real-Time Data:** Integrating live data from HR systems can improve model adaptability and responsiveness.
- **Advanced Deep Learning Models:** Exploring models such as LSTM or Transformer-based architectures could further improve predictive performance, especially in time-series attrition data.
- **Explainability and Interpretability:** Utilizing tools like SHAP or LIME to explain model predictions can help HR professionals better trust and understand the outcomes.
- **Handling Imbalanced Datasets:** Future work could apply advanced sampling techniques (e.g., SMOTE, ADASYN) or cost-sensitive learning to address class imbalance more effectively.
- **Cross-Industry Generalization:** Testing and adapting the models across various industries and geographic regions would help assess the generalizability of the findings.

Overall, this research lays a robust foundation for using machine learning in attrition prediction and opens pathways for more intelligent, data-driven HR management in the future.